

$U(1)$ -Symmetric Fermionic Gaussian Transfer-Matrix Formalism

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We organize the transfer-matrix formalism for number-conserving fermionic Gaussian operators in the notation of the handwritten derivation. A physical Gaussian state is labeled by the complex covariance $G = 2C - \mathbb{I}$, while a Gaussian operator is represented by a pure Choi covariance Σ on the doubled one-particle space $\mathcal{H}_L \oplus \mathcal{H}_R$. For a full-rank even Gaussian operator the doubled Choi correlation matrix gives an ordinary particle transfer matrix $T_p = C_{LL}^{-1}C_{LR}$, with the corresponding hole transfer matrix $T_h = (T_p^\dagger)^{-1}$. We then take the projective rank-one limits needed for occupation projectors and single-fermion gain/loss branches. The central correction is that branch-resolved single-fermion insertions remain pure Gaussian states, but their postselected Choi covariances are not determined by the singular effective transfer matrix alone. A right-leg creation or annihilation branch gives $T_{\text{eff}} = T_p(1 - P_\chi)$; the full singular datum also records the null-sector occupation, filled for $\hat{\chi}^\dagger$ and empty for $\hat{\chi}$.

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I. HAMILTONIAN-TO-COVARIANCE MAP

Let $c_i, c_i^\dagger$, $i = 1, \dots, N$, denote number-conserving fermion modes on the one-particle Hilbert space $\mathcal{H} \simeq \mathbb{C}^N$. For a single-particle Hamiltonian $h = h^\dagger$, write

$$h = u \operatorname{diag}(\epsilon_1, \dots, \epsilon_N) u^\dagger, \quad a_n^\dagger = \sum_i u_{in} c_i^\dagger. \quad (1)$$

The thermal Gaussian density matrix

$$\rho_h = \frac{1}{Z} \exp \left(- \sum_{ij} c_i^\dagger h_{ij} c_j \right) = \frac{1}{Z} \exp \left(- \sum_n \epsilon_n a_n^\dagger a_n \right) \quad (2)$$

has one-body correlation matrix

$$C_{ij} = \operatorname{Tr}(\rho_h c_i^\dagger c_j), \quad C = u f(h_{\text{diag}}) u^\dagger, \quad f(\epsilon) = \frac{1}{1 + e^\epsilon}. \quad (3)$$

The complex covariance used below is

$$G = 2C - \mathbb{I}. \quad (4)$$

In the zero-temperature or fixed-charge Slater limit C is an orthogonal projector and

$$G^2 = \mathbb{I}. \quad (5)$$

Equivalently, if the occupied orbitals form the columns of an isometry U_{occ} , then

$$|G\rangle = \prod_{\alpha=1}^q \left(\sum_i (U_{\text{occ}})_{i\alpha} c_i^\dagger \right) |0\rangle, \quad C = U_{\text{occ}} U_{\text{occ}}^\dagger, \quad G = 2U_{\text{occ}} U_{\text{occ}}^\dagger - \mathbb{I}. \quad (6)$$

This is the state-covariance convention of the first part of the handwritten notes.

II. PARTICLE AND HOLE EXCITATION RELATIONS

For a normalized one-particle mode

$$|\chi\rangle = \sum_i \chi_i |i\rangle, \quad \hat{\chi}^\dagger = \sum_i \chi_i c_i^\dagger, \quad \hat{\chi} = \sum_i \chi_i^* c_i, \quad P_\chi = |\chi\rangle\langle\chi|, \quad (7)$$

the only component that changes a fixed-charge Slater state is the component orthogonal to the already occupied or empty subspace. Thus, if the branch amplitude is nonzero, particle addition and particle removal update the physical state covariance by

$$C_{\text{add}} = C + \frac{(1-C)P_\chi(1-C)}{\text{Tr}[(1-C)P_\chi]}, \quad (8)$$

$$C_{\text{rem}} = C - \frac{CP_\chi C}{\text{Tr}[CP_\chi]}. \quad (9)$$

These equations are the fixed-charge state version of the Choi rank-one updates derived later. They also make clear which denominator is a Born probability: a creation branch requires an empty component of χ , while a removal branch requires an occupied component.

Let V be an invertible, number-conserving Gaussian operator. Up to a scalar it may be written as

$$V = \exp \left(\sum_{ij} c_i^\dagger A_{ij} c_j \right). \quad (10)$$

The particle and hole transfer matrices are defined by the conjugation rules

$$V c^\dagger(\phi) V^{-1} = c^\dagger(T_p \phi), \quad (11)$$

$$V c(\phi) V^{-1} = c(T_h \phi), \quad (12)$$

where $c^\dagger(\phi) = \sum_i \phi_i c_i^\dagger$ and $c(\phi) = \sum_i \phi_i^* c_i$. With the convention (10),

$$T_p = e^A, \quad T_h = (T_p^\dagger)^{-1}. \quad (13)$$

If $T_p = e^H u$ is the polar decomposition with $H = H^\dagger$ and u unitary, then

$$T_h = e^{-H} u. \quad (14)$$

This is the precise form of the particle/hole relation used in the transfer matrix part of the handwritten derivation.

III. FULL-RANK $U(1)$ GAUSSIAN OPERATORS

The full-rank even chart consists of operators generated by fermion bilinears, as in Eq. (10), with T_p invertible. Acting on a Slater determinant, V transports the occupied one-particle frame by T_p , followed by the Gram normalization required to return to a normalized Slater state. Thus the physical state transform is nonlinear in C , while the excitation transform is linear in the one-particle transfer matrix.

The two transfer matrices T_p and T_h are not independent. Equation (13) states that an invertible particle transfer matrix already fixes the hole transfer matrix. Conversely, if one begins from T_h , then

$$T_p = (T_h^\dagger)^{-1}. \quad (15)$$

The singular cases below should be understood as limits of this full-rank chart. That distinction is important: a finite full-rank T_p is a complete coordinate for an even Gaussian Choi covariance, but a singular effective matrix is not a complete coordinate for an odd postselected Choi covariance.

IV. CHOI VECTORIZATION AND DOUBLED COVARIANCE BLOCKS

Introduce two copies of the one-particle Hilbert space, $\mathcal{H}_L \oplus \mathcal{H}_R$, with fermion operators c_{Li} and c_{Ri} . The reference state is the number-conserving maximally entangled state

$$|\Omega\rangle_{LR} = 2^{-N/2} \prod_{i=1}^N (c_{Li}^\dagger + c_{Ri}^\dagger) |0\rangle_{LR}. \quad (16)$$

For one mode this is

$$|\Omega\rangle = \frac{|1_L 0_R\rangle + |0_L 1_R\rangle}{\sqrt{2}}. \quad (17)$$

We vectorize an operator by acting on the right leg,

$$|V\rangle\rangle = V_R |\Omega\rangle_{LR}. \quad (18)$$

The resulting Gaussian Choi state is labeled by its doubled correlation matrix and covariance

$$C_{\alpha\beta,ij} = \frac{\langle\langle V | c_{\alpha i}^\dagger c_{\beta j} | V \rangle\rangle}{\langle\langle V | V \rangle\rangle}, \quad \Sigma = 2C - \mathbb{I}_{L \oplus R}, \quad \alpha, \beta \in \{L, R\}. \quad (19)$$

We use the block notation

$$C = \begin{pmatrix} C_{LL} & C_{LR} \\ C_{RL} & C_{RR} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{RL} & \Sigma_{RR} \end{pmatrix}. \quad (20)$$

For a nonzero branch-resolved Gaussian Choi vector,

$$C^2 = C, \quad \Sigma^2 = \mathbb{I}_{L \oplus R}. \quad (21)$$

V. FULL-RANK TRANSFER-MATRIX/CHOI DICTIONARY

For an even full-rank Gaussian operator with particle transfer matrix $T \equiv T_p$, define

$$L = (\mathbb{I} + TT^\dagger)^{-1}, \quad R = (\mathbb{I} + T^\dagger T)^{-1}. \quad (22)$$

The Choi correlation blocks are

$$C_{LL} = L, \quad C_{LR} = LT = TR, \quad (23)$$

$$C_{RL} = T^\dagger L = RT^\dagger, \quad C_{RR} = T^\dagger LT = \mathbb{I} - R. \quad (24)$$

Equivalently, in covariance variables,

$$\Sigma_{LL} = 2L - \mathbb{I}, \quad \Sigma_{LR} = 2LT, \quad (25)$$

$$\Sigma_{RL} = 2T^\dagger L, \quad \Sigma_{RR} = \mathbb{I} - 2R. \quad (26)$$

The inverse map in the full-rank chart is

$$T = C_{LL}^{-1} C_{LR} = (\mathbb{I} + \Sigma_{LL})^{-1} \Sigma_{LR}. \quad (27)$$

The second equality follows from $\mathbb{I} + \Sigma_{LL} = 2C_{LL}$ and $\Sigma_{LR} = 2C_{LR}$. Equation (27) is the formula that must be used only on the support where the even full-rank chart is valid.

Proposition 1. *The blocks (23) and (24) define a pure Choi state: $C^2 = C$, or equivalently $\Sigma^2 = \mathbb{I}$.*

Proof. The identities $LT = TR$, $LTT^\dagger = \mathbb{I} - L$, and $RT^\dagger T = \mathbb{I} - R$ imply

$$C_{LL}^2 + C_{LR}C_{RL} = L^2 + LTT^\dagger L = L, \quad (28)$$

$$C_{LL}C_{LR} + C_{LR}C_{RR} = L^2T + LT(\mathbb{I} - R) = LT, \quad (29)$$

with the remaining two block equations following by Hermitian conjugation and the same R -identity. $\square$

VI. RANK-ONE PROJECTORS AND PROJECTIVE LIMITS

Let $P = P_\chi = |\chi\rangle\langle\chi|$ and $Q = \mathbb{I} - P$. The occupation projectors $N_\chi = \hat{\chi}^\dagger \hat{\chi}$ and $1 - N_\chi$ arise as projective limits of full-rank even Gaussian operators:

$$N_\chi = \lim_{\lambda \rightarrow \infty} e^{-\lambda} e^{\lambda N_\chi}, \quad T_\lambda^{(N)} = Q + e^\lambda P, \quad (30)$$

$$1 - N_\chi = \lim_{\lambda \rightarrow \infty} e^{-\lambda N_\chi}, \quad T_\lambda^{(1-N)} = Q + e^{-\lambda} P. \quad (31)$$

At the transfer-matrix level the first limit diverges on $\text{Ran } P$, while the second becomes rank deficient. At the Choi-covariance level both limits are finite. On the P -sector,

$$C(T = t) = \frac{1}{1 + |t|^2} \begin{pmatrix} 1 & t \\ t^* & |t|^2 \end{pmatrix}_{LR}. \quad (32)$$

Thus $t \rightarrow \infty$ gives $|0_L 1_R\rangle$, whereas $t \rightarrow 0$ gives $|1_L 0_R\rangle$. These are the two even null completions of the full-rank chart. They are not the odd single-fermion gain/loss completions discussed below.

VII. SINGLE-FERMION GAIN AND LOSS BRANCHES

Now let V be an even full-rank Gaussian operator and consider the branch-resolved odd Kraus amplitudes

$$K_+ = \hat{\chi}^\dagger V, \quad K_- = \hat{\chi} V. \quad (33)$$

Because $K_\pm$ are odd, they do not define ordinary invertible even conjugation maps. The correct object is instead the Choi vector

$$|K_+\rangle\rangle = \hat{\chi}_R^\dagger |V\rangle\rangle, \quad |K_-\rangle\rangle = \hat{\chi}_R |V\rangle\rangle. \quad (34)$$

If the corresponding branch amplitude is nonzero, these are still pure Gaussian Choi states. Therefore the updated covariance still satisfies

$$\Sigma_\pm^2 = \mathbb{I}. \quad (35)$$

The failure is not purity. The failure is the assumption that an odd postselected Choi state can be reconstructed from the ordinary full-rank even transfer coordinate alone.

Set

$$u_R = \begin{pmatrix} 0 \\ |\chi\rangle \end{pmatrix}_{L \oplus R}. \quad (36)$$

Creation fills the empty projection of u_R , while annihilation removes its occupied projection:

$$C_+ = C + \frac{(\mathbb{I} - C)u_R u_R^\dagger (\mathbb{I} - C)}{u_R^\dagger (\mathbb{I} - C)u_R}, \quad (37)$$

$$C_- = C - \frac{C u_R u_R^\dagger C}{u_R^\dagger C u_R}. \quad (38)$$

The denominators

$$d_+ = \langle \chi | (\mathbb{I} - C_{RR}) | \chi \rangle = \text{Tr}[(\mathbb{I} - C_{RR})P], \quad d_- = \langle \chi | C_{RR} | \chi \rangle = \text{Tr}[C_{RR}P] \quad (39)$$

are the branch probabilities in the normalized Choi state. The formulas are defined for $d_\pm \neq 0$.

Blockwise, the creation branch is

$$C_{LL}^{(+)} = C_{LL} + \frac{C_{LR} P C_{RL}}{d_+}, \quad (40)$$

$$C_{LR}^{(+)} = C_{LR} - \frac{C_{LR} P (\mathbb{I} - C_{RR})}{d_+}, \quad (41)$$

$$C_{RR}^{(+)} = C_{RR} + \frac{(\mathbb{I} - C_{RR}) P (\mathbb{I} - C_{RR})}{d_+}, \quad (42)$$

with $C_{RL}^{(+)} = (C_{LR}^{(+)})^\dagger$. The loss branch is

$$C_{LL}^{(-)} = C_{LL} - \frac{C_{LR} P C_{RL}}{d_-}, \quad (43)$$

$$C_{LR}^{(-)} = C_{LR} - \frac{C_{LR} P C_{RR}}{d_-}, \quad (44)$$

$$C_{RR}^{(-)} = C_{RR} - \frac{C_{RR} P C_{RR}}{d_-}, \quad (45)$$

with $C_{RL}^{(-)} = (C_{LR}^{(-)})^\dagger$.

VIII. EFFECTIVE SINGULAR TRANSFER MATRIX

Suppose the pre-insertion Choi covariance is in the full-rank even chart (23)–(24), with $T = T_p$. After the right-leg insertion the ordinary inverse in (27) must be replaced by the support inverse, or equivalently by a full-rank regularization followed by the singular limit. Substituting Eqs. (23)–(24) into the rank-one updates gives

$$T_{\text{eff}}^{(+)} = (C_{LL}^{(+)})_{\text{supp}}^{-1} C_{LR}^{(+)} = TQ, \quad Q = \mathbb{I} - P, \quad (46)$$

and similarly

$$T_{\text{eff}}^{(-)} = (C_{LL}^{(-)})_{\text{supp}}^{-1} C_{LR}^{(-)} = TQ. \quad (47)$$

Thus a right-leg single-fermion insertion removes the χ -column direction of the particle transfer matrix:

$$T_p \mapsto T_p (\mathbb{I} - P_\chi). \quad (48)$$

The analogous left-leg insertion removes the corresponding row direction:

$$T_p \mapsto (\mathbb{I} - P_\chi) T_p. \quad (49)$$

Remark 1 (First corrected pitfall: regulating the final odd covariance). One may regulate the parent even operator or use the exact rank-one update (37) or (38). What is not legitimate is to interpolate the final odd covariance by hand and then continue to impose pure-state transfer identities. For example,

$$\Sigma_\varepsilon = \begin{pmatrix} P + \varepsilon Q & Q + \varepsilon P \\ Q + \varepsilon P & P + \varepsilon Q \end{pmatrix} \quad (50)$$

has, on the P -sector,

$$\Sigma_\varepsilon|_P = \begin{pmatrix} 1 & \varepsilon \\ \varepsilon & 1 \end{pmatrix}, \quad \Sigma_\varepsilon^2|_P = \begin{pmatrix} 1 + \varepsilon^2 & 2\varepsilon \\ 2\varepsilon & 1 + \varepsilon^2 \end{pmatrix} \neq \mathbb{I}. \quad (51)$$

Hence such a finite- ε object is not a pure Gaussian Choi covariance.

IX. NULL-SECTOR OCCUPATION DATA

The effective matrix $T_{\text{eff}} = TQ$ captures the homogeneous propagation sector, but it does not determine the full postselected Choi covariance. It only says that the right-leg P -column no longer participates in the L -to- R bridge. It does not say whether that disconnected mode is empty or filled.

The distinction is already visible for one mode with $T = 1$. The reference state is Eq. (17). Right-leg creation gives

$$\hat{\chi}_R^\dagger|\Omega\rangle \propto |1_L 1_R\rangle, \quad C_+ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{LR}, \quad (52)$$

whereas right-leg loss gives

$$\hat{\chi}_R|\Omega\rangle \propto |0_L 0_R\rangle, \quad C_- = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}_{LR}. \quad (53)$$

Both branches have $T_{\text{eff}} = 0$. They are nevertheless different pure Choi states.

Equivalently, reconstructing from $T_{\text{eff}} = 0$ using only the even formula (32) gives one of the even null completions:

$$t \rightarrow 0: \quad C(t) \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}_{LR} \quad (|1_L 0_R\rangle), \quad (54)$$

or

$$t \rightarrow \infty: \quad C(t) \rightarrow \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}_{LR} \quad (|0_L 1_R\rangle). \quad (55)$$

Neither is the odd creation completion $|1_L 1_R\rangle$, and neither is the odd loss completion $|0_L 0_R\rangle$.

Remark 2 (Second corrected pitfall: regularizing T_{eff} alone). A singular effective transfer matrix can be regularized as an even matrix, but that regularization chooses an even null completion. It cannot recover which odd branch occurred. The missing information is precisely the null-sector occupation.

The complete singular data for a right-leg odd branch are therefore

$$(T_{\text{eff}}, P_{\text{null}}, n_{\text{null}}), \quad P_{\text{null}} = P_\chi, \quad (56)$$

where the occupation datum may be read from the right-right Choi block:

$$P_\chi C_{RR}^{(+)} P_\chi = P_\chi, \quad P_\chi C_{RR}^{(-)} P_\chi = 0. \quad (57)$$

Thus

$$n_{\text{null}} = \begin{cases} 1, & \text{right-leg creation } \hat{\chi}_R^\dagger, \\ 0, & \text{right-leg loss } \hat{\chi}_R. \end{cases} \quad (58)$$

X. FINAL PICTURE

The full-rank even formalism is internally consistent and complete:

$$T_p \iff C(T_p) \iff \Sigma(T_p), \quad T_h = (T_p^\dagger)^{-1}. \quad (59)$$

Rank-one even projectors lie on the boundary of this chart: the transfer matrix may diverge or lose rank, but the Choi covariance has a finite projective limit. Odd single-fermion branches go one step further. They are valid pure Gaussian Choi states, and their covariances must be computed by the rank-one creation/loss updates (37) and (38). The extracted singular transfer matrix is

$$T_{\text{eff}} = T_p(\mathbb{I} - P_\chi) \quad (60)$$

for a right-leg insertion, but this matrix is only the propagation part of the answer. The full postselected Choi covariance also requires the null-sector occupation in Eq. (57). This resolves the apparent contradiction in the handwritten derivation: purity is preserved, the Choi representation remains valid, and the effective transfer matrix remains useful, but the singular odd branch is not coordinatized by T_{eff} alone.